

Concentration of Benzoquinone as a Result of Catechol Oxidation at Varying Volumes of Catechol in *Solanum tubersum*

ABSTRACT

In this experiment of the oxidation of catechol, the goal was to identify how varying the volumes of the substrate catechol when mixed with the enzyme catecholase, obtained from a sample of *Solanum tubersum*, would impact the concentration of the oxidation product benzoquinone. The prediction is that as the volume of catechol increases, the resulting concentration of benzoquinone will increase, establishing the independent variable as catechol volume and the dependent variable as concentration of benzoquinone. Three treatments were created, the first with 2.5 mL catechol to 1.5 mL water, the second with 1.25 mL catechol to 2.75 mL water, and the third with 0.625 mL catechol to 3.375 mL water. Each treatment also contained 500 μ L of catecholase to produce an enzymatic reaction. For each treatment group, there were three replicates to increase accuracy, as well as a blank treatment consisting of 4 mL of water and 500 μ L of catecholase to calibrate the program. With these treatments, absorbance was observed to calculate the resulting concentration of benzoquinone with the JMP program. A general trend was found that greater volumes of catechol resulted in a greater concentration of benzoquinone, however, a large standard of error for the 2.5 mL catechol treatment skews the results. Due to human error, the presence of a significant difference cannot be determined, and thus I fail to reject the null hypothesis. However, the existing trend proves the prediction correct and provides insight into enzymatic reactions. More specifically, the process of catechol oxidation details the preservation of fruit from harmful bacteria through enzymatic browning.

INTRODUCTION

In cellular processes, benzoquinone is the result of the oxidation of catechol, which is usually released when damage is done to plant cells, such as cutting an apple or peeling a potato. When exposed to oxygen, catechol oxidizes with the oxidizing agent catecholase and produces benzoquinone, which is recognized as the browning of the fruit or vegetable. Benzoquinone is a product that protects the fruit or vegetable after being cut. Without it, the fruit or vegetable is susceptible to further damage by bacteria. However, this process does not occur fast enough on its own. Because most biological processes do not happen as fast as they need to, enzymes assist in ensuring they occur in the necessary timeframe. Enzymes speed up reactions by lowering the amount of energy needed for the reactions to complete, so that they can occur more quickly. Enzymes have an active site, in which a substrate attaches and the process of making products is sped up and released back to the cell more quickly than if not assisted by an enzyme (Gregg, et al., 2023). In the case of catechol oxidation, the substrate catechol attaches to the enzyme catecholase, and produces benzoquinone.

The goal of this experiment was to determine the effect of varying substrate volumes of catechol on the resulting concentration of the product benzoquinone. In a study done by Dawson and Nelson, it was concluded that the concentration of benzoquinone was directly related to the volume of catechol that reacts (Dawson & Nelson, 1938, p. 249). This study was based off another study by Fieser and Peters, who believed there was a trend resulted in their data when observing different ratios of catechol with catecholase on the concentration of benzoquinone (Fieser & Peters, 1931, pp. 802-803). Given this information, the prediction is that an increased volume of catechol will result in a greater volume of benzoquinone. With volume of catechol as the independent variable and production of benzoquinone as the dependent variable, the alternative hypothesis is that an increased volume of catechol in the solution would produce a greater concentration of benzoquinone, and a decreased volume of catechol would produce a lesser concentration of benzoquinone. The null hypothesis is that in the oxidation of catechol, there will be no significant difference in the resulting concentration of benzoquinone among the three treatments with varying volumes of the substrate catechol.

METHODS

DATA COLLECTION

To test the effect of the catechol substrate volume on the resulting benzoquinone, three separate treatments were created, each with three replicates, consisting of catecholase, which was extracted from *Solanum tuberosum*, water, and varying volumes of catechol that decreased at an even rate to distinguish a significant difference. The independent variable is the volume of catechol in mL, the dependent variable is the absorbance value of benzoquinone in ODU, and for each treatment, a blank was created with no catechol present. The values of water, catechol, and catecholase for each treatment are listed in Table 3-2. Each treatment was observed in spectrometer, with the blank used to calibrate the program, and the absorbance values for the replicates of each treatment were collected at a wavelength of 500 nm over 100 seconds (Gregg, et al., 2023).

Table 3-2. Solutions for treatments A, B, and C, with varying measures of water, catechol, and catecholase. Each treatment had 3 replicates, with 500 microliters of catecholase and different variations of water and catechol that equal 4 mL.

Solution	Blank		Treatment (3 replicates each)		
	Water	Catecholase	Water	Catechol	Catecholase
2.5 ml Catechol (Treatment A)	4 mL	500 µL	1.5 mL	2.5 mL	500 µL
1.25 mL Catechol (Treatment B)	4 mL	500 µL	2.75 mL	1.25 mL	500 µL
0.625 mL Catechol (Treatment C)	4 mL	500 µL	3.375 mL	0.75 mL	500 µL

DATA ANALYSIS

To determine the concentration of benzoquinone, the mean absorbance of the three replicates for each treatment was found, then used to determine mean concentration in mM with the standard curve equation of $y = 1.0905x + 0.0014$ (Gregg, et al., 2023). This was done by taking the mean absorbance (ODU) and plugging that value in for y, then solving for x. The absorbance values for each treatment were plugged into an excel spreadsheet which provided the corresponding values of concentration over the 100 seconds in 20 second intervals. This sheet was exported into JMP, then the builder tool was used to configure a scatterplot. The scatterplot depicts the concentration of benzoquinone (y-axis) across time in 20 second intervals (x-axis). To create the line of best fit, the mean and standard deviations were selected on the program; this data is demonstrated in Figure 1-1. Then to create the bar graph, the values except for when time was equal to 100 seconds were selected and excluded from the data set. The graph builder tool was used again to create a bar graph which depicts benzoquinone concentration (y-axis) that resulted from varying volumes of catechol (x-axis) at the 100 second mark. Standard deviation bars were created for this graph, and this data is represented in Figure 1-2. To test if the data was significantly different, the Analysis of Variance (ANOVA) test was performed, however, a t-test was not run (Gregg, et al., 2023). This data was collected in table 4-1.

RESULTS

All 3 treatments (2.5 mL catechol, 1.25 mL catechol, and 0.625 mL catechol) demonstrate an increase of the concentration of benzoquinone over the 100 seconds at the 500 nm wavelength. With catechol volume as the independent variable and concentration of benzoquinone as the dependent variable, a general trend follows that an increased volume of catechol produced a greater concentration of benzoquinone. However, a large standard deviation for the 2.5 mL treatment defies the prediction. It is evident that the treatment with 1.25 mL catechol

has a higher concentration of benzoquinone than the treatment with 2.5 mL catechol across one hundred seconds, which goes against the alternative hypothesis (Figure 1-1). This trend presents no significant difference in the data of these two treatments. The treatment of 0.625 mL catechol demonstrates a lower concentration of benzoquinone which correlates with a decrease in the volume of the catechol substrate (Figure 1-1). However, the calculated ANOVA p-value of 0.0564 determines that there is no significant difference across all the treatment sets (Table 4-1). Figure 1-2 depicts a larger standard of error for the 2.5 mL catechol treatment and 0.625 mL catechol treatment compared to the 1.25 mL catechol treatment, accounting for an overall lack of significant difference in the treatments. The t-test results were not included in Figure 1-2 due to no significant difference across the treatments.

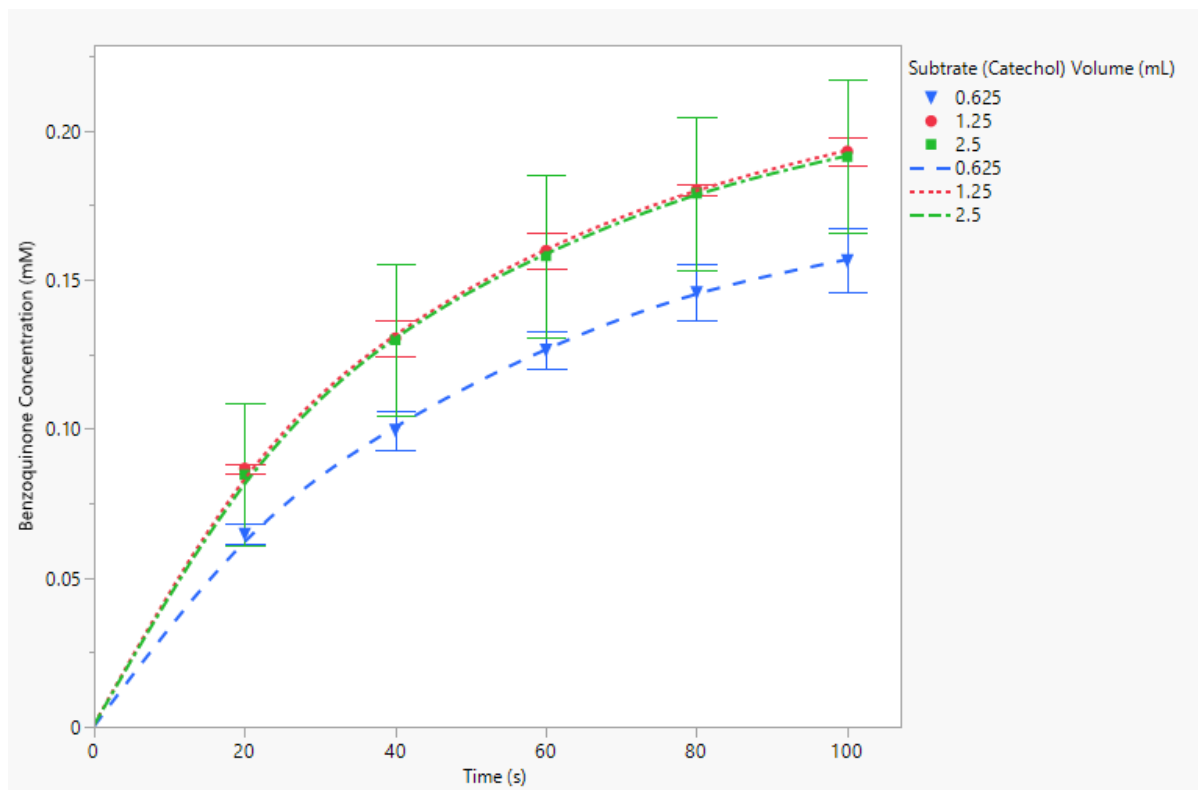


Figure 1-1. Concentration of benzoquinone in mM, calculated by observing wavelength at 500 nm, for 3 treatments with varying volumes of the substrate catechol, in mL, recorded across 100 seconds. The lines represent a line of best fit, or mean of the data, which minimizes errors from the collected data, and the bars represent the mean $\pm$ standard error for 3 replicates of each treatment.

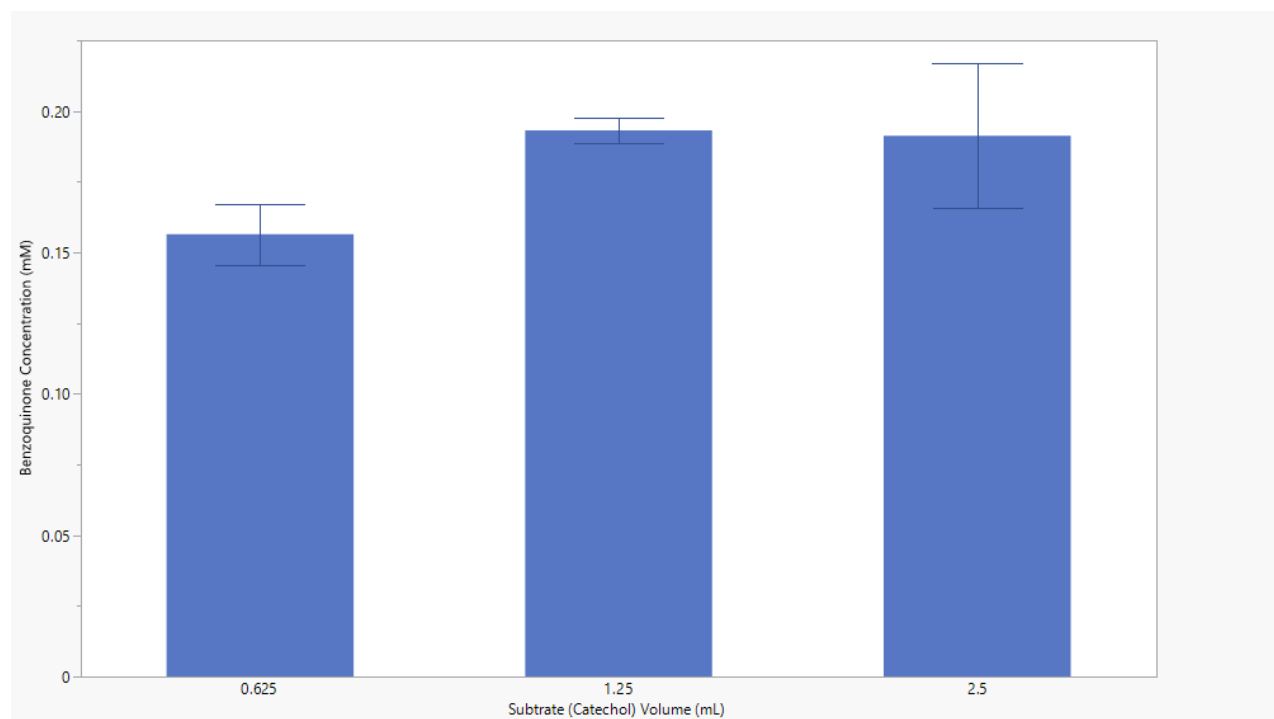


Figure 1-2. Concentration of benzoquinone in mM, calculated by observing wavelength at 500 nm, for 3 treatments with varying volumes of the substrate catechol (0.625, 1.25, and 2.5) in mL. The values demonstrate the mean $\pm$ standard error from 3 replicates of each treatment. The ANOVA p-value is .0564.

Table 4-1. Mean, standard deviation, ANOVA p-Value, and t-test letters for 3 treatments of varying volumes of the substrate catechol in mL. The volumes of catechol in the treatments are 0.625 mL, 1.25 mL, and 2.5 mL, and the mean and standard deviation are derived from the concentration values of benzoquinone over 100 seconds.

Treatment	Mean	Standard Deviation	ANOVA p-Value	t-Test Letters
.625	.156442	.0106927	.0564	A
1.25	.1931224	.0045092		A
2.5	.1912884	.0255016		B

DISCUSSION

Based on the results, there was no statistical impact on the concentration of benzoquinone when the volumes of catechol were varied. However, a general trend of higher catechol concentration results in a higher concentration of benzoquinone, which is because the greater amount of a reacting agent present will produce a greater resulting concentration (Dawson & Nelson, 1938, p. 249). The alternative hypothesis, which stated that in the oxidation of catechol, a greater volume of catechol will have a significant effect on the resulting concentration of benzoquinone, was incorrect. Since the p-value was greater than 0.05 ($p=0.0564$), I fail to reject the null hypothesis because there is not enough evidence to say the greater volumes of catechol had a significant difference on the resulting concentration of benzoquinone.

The lack of significance among treatments may be due to human error since the general trends support the prediction. It is evident that the 2.5 mL treatment has a larger standard of error than the other treatment groups, which is likely caused from this treatment being the first one tested, and explains why this is the area that the prediction fails to support. Thus, it could be possible that with further testing, the null hypothesis can be rejected. However, not enough information from this experiment is available to confirm this. Future testing could also include larger volumes of catechol, all evenly spaced out, so that a resulting concentration of benzoquinone may be more evident in the results. Overall, the research presents vital information regarding the

preservation of fruits and vegetables, and can be used in various fields, from harvesting to food markets, to maintain the quality of produce more effectively.

REFERENCE LIST

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